

6.3 Transition Element Complexes: Isomers, Reactions & Stability (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	6. Inorganic Chemistry (A Level Only)
Topic	6.3 Transition Element Complexes: Isomers, Reactions & Stability (A Level Only)
Difficulty	Medium

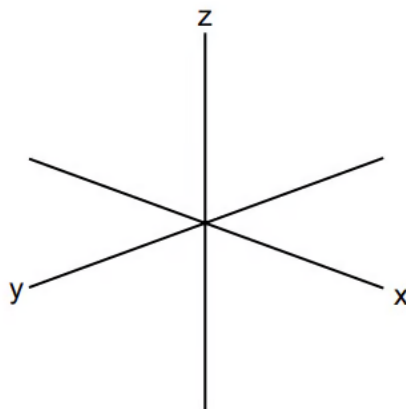
Time allowed: 70

Score: /58

Percentage: /100

Question 1a

Sketch the shape of a $3d_{xy}$ orbital.



[1 mark]

Question 1b

i)

Some transition elements and their compounds behave as catalysts. Explain why transition elements behave as catalysts.

[2]

ii)

Catalysis can be classified as heterogeneous or homogeneous.

Complete Table 7.1 by placing **one** tick (✓) in each row to indicate the type of catalysis in each reaction.

Table 7.1

reaction	type of catalysis	
	heterogeneous	homogeneous
Fe in the Haber process		
Fe^{2+} in the $\text{I}^- / \text{S}_2\text{O}_8^{2-}$ reaction		
NO_2 in the oxidation of SO_2		

[1]

[3 marks]

Question 1c

A solution containing a mixture of $\text{Sn}^{2+}(\text{aq})$ and $\text{Sn}^{4+}(\text{aq})$ is added to a solution containing a mixture of $\text{Fe}^{2+}(\text{aq})$ and $\text{Fe}^{3+}(\text{aq})$.

Table 7.2 lists electrode potentials for some electrode reactions of these ions.

Table 7.2

electrode reaction	$E^\ominus / \text{V}$
$\text{Fe}^{2+} + 2\text{e}^- \rightleftharpoons \text{Fe}$	-0.44
$\text{Fe}^{3+} + 3\text{e}^- \rightleftharpoons \text{Fe}$	-0.04
$\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+}$	+0.77
$\text{Sn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Sn}$	-0.14
$\text{Sn}^{4+} + 2\text{e}^- \rightleftharpoons \text{Sn}^{2+}$	+0.15

$E^\ominus$ data from the table can be used to predict the reaction that takes place when the two solutions are mixed.

i)

Write an equation for this reaction.

[1]

ii)

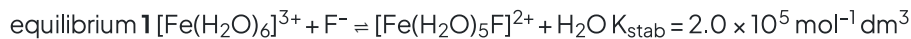
Calculate $E^\ominus_{\text{cell}}$ for this reaction.

[1]

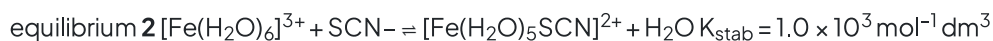
[2 marks]

Question 1d

Hexaaquairon(III) ions are pale violet. They form a colourless complex with fluoride ions, F^- , as shown in equilibrium **1**, and a deep-red complex with thiocyanate ions, SCN^- , as shown in equilibrium **2**.



violet colourless



violet deep-red

The following two experiments are carried out.

Experiment 1: A few drops of $\text{KSCN}(\text{aq})$ are added to 5 cm^3 of $\text{Fe}^{3+}(\text{aq})$, followed by a few drops of $\text{KF}(\text{aq})$.

Experiment 2: A few drops of $\text{KF}(\text{aq})$ are added to 5 cm^3 of $\text{Fe}^{3+}(\text{aq})$, followed by a few drops of $\text{KSCN}(\text{aq})$.

i)

Predict and explain the sequence of colour changes you would observe in each of Experiment 1 and Experiment 2.

Experiment 1

Experiment 2

[4]

ii)

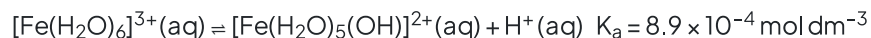
Name the type of reaction occurring during the experiments in **(d)(i)**.

[1]

[5 marks]

Question 1e

Solutions of iron(III) salts are acidic due to the equilibrium shown.



Calculate the pH of a 0.25 mol dm^{-3} FeCl_3 solution.

Show your working.

pH =

[2 marks]

Question 2a

This question is about transition metal complexes.

An aqueous solution of copper(II) contains the $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ complex ion.

Define the term complex ion.

[1 mark]

Question 2b

A sample of $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ is reacted with aqueous sodium hydroxide and a separate sample of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ with an excess of aqueous ammonia.

Give the following information about these reactions.

- the reaction of $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ with aqueous sodium hydroxide

colour and state of the copper-containing species

.....

ionic equation

type of reaction

- the reaction of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ with an excess of aqueous ammonia

colour and state of the cobalt-containing species

.....

ionic equation

type of reaction

[6 marks]

Question 2c

The $[\text{Fe}(\text{NH}_3)_2\text{CN}_4]^-$ complex shows stereoisomerism.

Complete the three-dimensional diagrams to show the two isomers of $[\text{Fe}(\text{NH}_3)_2\text{CN}_4]^-$.

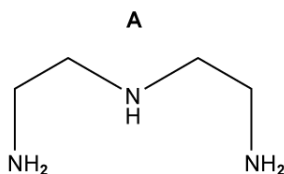


Suggest the type of stereoisomers.

[3 marks]

Question 2d

Compound **A**, $C_4H_{13}N_3$, is a tridentate ligand.



- i)
Suggest why one molecule of **A** can form three dative bonds.
- ii)
 $C_4H_{13}N_3$ reacts with aqueous chromium(III) ions, $[Cr(H_2O)_6]^{3+}$, in a 2:1 ratio to form a new complex ion.

Construct an equation for this reaction.

[1]

[1]

[2 marks]

Question 3a

Cobalt is a transition element with atomic number 27.

Complete the electronic configurations of a Co atom and a Co^{2+} ion.

Co atom $1s^2 2s^2 2p^6$

Co^{2+} ion $1s^2 2s^2 2p^6$

[2 marks]

Question 3b

Co^{2+} ions form a linear complex with Cl^- ions, which are monodentate ligands.

Draw the structure of this complex and include its overall charge.

structure

charge

[2 marks]

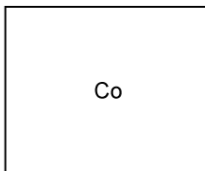
Question 3c

Co^{2+} ions exist as $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ complex ions in an aqueous solution.

Complete a three-dimensional diagram to show the shape of this complex.

Name its shape.

Label and state the value of two bond angles.

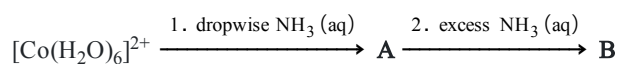


name of shape

[3 marks]

Question 3d

When NH_3 (aq) is added to Co^{2+} (aq), dropwise at first and then in excess, two chemical reactions occur as shown.



For each reaction, describe what you would see and write an equation.

reaction 1

observation

equation

reaction 2

observation

equation

[4 marks]

Question 3e

EDTA⁴⁻ is a hexadentate ligand. When a solution of EDTA⁴⁻ is added to a solution containing [Co(H₂O)₆]²⁺ a new complex is formed. The formula of this complex is [CoEDTA]²⁻.

i)

Explain what is meant by the term hexadentate ligand.

[2]

ii)

Name the type of reaction occurring here.

[1]

iii)

Write an expression for the stability constant, K_{stab} , of [CoEDTA]²⁻ in this reaction.

[1]

iv)

The numerical value of the K_{stab} of [CoEDTA]²⁻ is $7.7 \times 10^4 \text{ mol}^{-6} \text{ dm}^3$ at 298K.

State what this tells us about the [CoEDTA]²⁻ complex ion.

[1]

[5 marks]

Question 4a

The chloride ion is a monodentate ligand. When concentrated hydrochloric acid is added to $[\text{Co}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$ ions the water ligands are replaced.

- i)
Suggest an equation for the reaction of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$ with concentrated hydrochloric acid.

[2]

- iii)
Draw the three-dimensional diagram to show the complex formed in the reaction between $[\text{Co}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$ and HCl.

[2]

[4 marks]**Question 4b**

$[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ is an example of an octahedral complex which exhibits cis-trans isomerism.

- i)
Deduce the oxidation state of cobalt in this complex. Explain your answer.

[2]

- ii)
Draw structures of the two $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ isomers clearly identifying each isomer.

[3]

[5 marks]

Question 4c

Explain the origin of colour in a transition element complex such as $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$.

[3 marks]

Question 4d

Bidentate ligands form two coordinate bonds to a central metal atom or ion in a complex. 1,2-diaminoethane is an example of a compound which is a bidentate ligand. When added to $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ all of the other ligands are replaced and two optical isomers are formed.

i)

Suggest an equation for the reaction of $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ with 1,2-diaminoethane.

[2]

ii)

Draw the two isomeric forms of the product of the reaction of $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ with 1,2-diaminoethane.

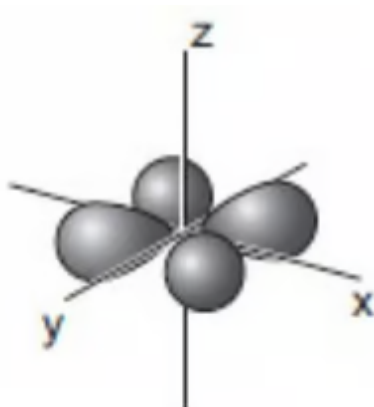
[3]

[5 marks]

Model Answers: Medium

1a

a) The shape of a $3d_{xy}$ orbital:



- Correct shape; [1 mark]

[Total: 1 mark]

- Drawing the shapes takes some practice, but the key thing is to have four lobes or two figures of eight that lie in the same plane at right angles to each other
- Note that the lobes fall between the x and y planes, not along the axes

1b

b)

i) The reason why transition elements behave as catalysts is:

- Transition elements have more than one stable oxidation state; [1 mark]
- There are vacant orbitals that are energetically accessible

OR

They can form dative bonds with ligands; [1 mark]

ii) The type of catalysis in each reaction are:

	type of catalysis
--	-------------------

reaction	heterogeneous	homogeneous
Fe in the Haber process	✓	
Fe^{2+} in the $\text{I}^- / \text{S}_2\text{O}_8^{2-}$ reaction		✓
NO_2 in the oxidation of SO_2		✓

- All **three** correct; [1 mark]

[Total: 3 marks]

- A heterogeneous catalyst is in a different phase from the reactants
 - The Haber process uses gaseous reactants and the iron is solid so it is a heterogeneous catalyst
- Homogeneous catalysts are in the same phase
 - Fe^{2+} in the I^- and $\text{S}_2\text{O}_8^{2-}$ are all aqueous ions
 - NO_2 and SO_2 are both gases

1c

c)

i) An equation for this reaction is:

- $\text{Sn}^{2+} + 2\text{Fe}^{3+} \rightarrow \text{Sn}^{4+} + 2\text{Fe}^{2+}$; [1 mark]

ii) The value of $E^\ominus_{\text{cell}}$ for this reaction is:

- 0.62 (V); [1 mark]

[Total: 2 marks]

- The $\text{Sn}^{2+} / \text{Sn}^{4+}$ electrode is more negative than the $\text{Fe}^{2+} / \text{Fe}^{3+}$ electrode
 - This means it will act as a negative electrode and go 'backwards' releasing electrons
 - The Sn^{2+} will be the reactant
- The $\text{Fe}^{2+} / \text{Fe}^{3+}$ will go in the forward direction
 - So, Fe^{2+} is the reactant
- The $\text{Fe}^{2+} / \text{Fe}^{3+}$ electrode must be doubled so that 2 electrons transfer from one electrode to the other
- To calculate $E^\ominus_{\text{cell}}$ use the formula

$$E^\ominus_{\text{cell}} = E^\ominus_{\text{right}} - E^\ominus_{\text{left}}$$

- By convention, the left electrode is the more negative electrode where oxidation takes place



- You can also use the alternative convention:

$$E_{\text{cell}}^{\ominus} = E_{\text{reduction}}^{\ominus} - E_{\text{oxidation}}^{\ominus}$$

- Either way will get you to the right answer

$$E_{\text{cell}}^{\ominus} = 0.77 - (0.15) = 0.62$$

1d

d)

i) Predicting and explaining the sequence of colour changes you would observe in each of Experiment 1 and Experiment 2:

Predictions (max 3 marks)

Experiment 1:

Violet → deep-red

Deep-red → colourless

Experiment 2:

Violet → colourless

Which stays colourless / does not change

- 2 observations; [1 mark]
- 3 observations; [2 marks]
- 4 observations; [3 marks]

Explanation:

- $K_{\text{stab}} [\text{Fe}(\text{H}_2\text{O})_5\text{F}]^{2+} > [\text{Fe}(\text{H}_2\text{O})_5\text{SCN}]^{2+}$

OR

K_{stab} for eqm 1 is greater than eqm 2; [1 mark]

ii) The type of reaction occurring during the experiments in (d)(i) is:

- Ligand displacement / exchange / substitution ; [1 mark]

[Total: 5 marks]

- The larger the stability constant, the more stable a complex ion is
- By comparing the two values you can see that the complex ion in the first reaction is more stable than the complex ion in the second reaction
 - $2.0 \times 10^5 > 1.0 \times 10^3$
- Adding KSCN (aq) to violet Fe^{3+} (aq) forms the deep-red $[\text{Fe}(\text{H}_2\text{O})_5\text{SCN}]^{2+}$
 - This complex is displaced by the addition of KF, since the resulting colourless $[\text{Fe}(\text{H}_2\text{O})_5\text{F}]^{2+}$ complex is more stable
- In experiment 2, adding KF (aq) to violet Fe^{3+} (aq), enables the formation of the colourless $[\text{Fe}(\text{H}_2\text{O})_5\text{F}]^{2+}$ complex
 - This complex is not displaced by the addition of KSCN (aq) so it remains colourless

1e

e) The pH of a $0.25 \text{ mol dm}^{-3} \text{ FeCl}_3$ solution:

- $[\text{H}^+]^2 = 8.9 \times 10^{-4} \times 0.25$
AND
 $[\text{H}^+] = 0.0149$; [1 mark]
- $\text{pH} = -\log(0.0149) = 1.8$; [1 mark]

[Total: 2 marks]

- To solve this problem you are using a rearrangement of
$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$
- There are equal numbers of H^+ and A^- ions so the expression simplifies to
$$[\text{H}^+]^2 = K_a \times [\text{HA}]$$
- An assumption is made that the equilibrium concentration of HA is the same as the initial concentration since very little will have dissociated
- The square root is taken and then the pH is found using $\text{pH} = -\log [\text{H}^+]$

2a

a)

i) A complex ion is:

- A molecule / ion formed by a (central) metal atom / ion being surrounded by / bonded to (one or more) ligands; [1 mark]

[Total: 1 mark]

- You are expected to know the definition of a complex ion

2b

b) The complete information for the reactions is:

The reaction of $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ with aqueous sodium hydroxide:

- Colour and state of the copper-containing species = Blue precipitate / solid; [1 mark]
- Ionic equation = $[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 2\text{OH}^- \rightarrow \text{Cu}(\text{OH})_2 + 6\text{H}_2\text{O}$
OR
 $[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 2\text{OH}^- \rightarrow [\text{Cu}(\text{H}_2\text{O})_4(\text{OH})_2] + 2\text{H}_2\text{O}$; [1 mark]
- Type of reaction = Precipitation
OR
Acid-base; [1 mark]

The reaction of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ with an excess of aqueous ammonia:

- Colour and state of the cobalt-containing species = (Straw) yellow / brown solution; [1 mark]
- Ionic equation = $[\text{Co}(\text{H}_2\text{O})_6]^{2+} + 6\text{NH}_3 \rightarrow [\text{Co}(\text{NH}_3)_6]^{2+} + 6\text{H}_2\text{O}$; [1 mark]
- Type of reaction = ligand exchange / ligand displacement / ligand substitution / ligand replacement; [1 mark]

[Total: 6 marks]

- M^{2+} complex ions react with aqueous sodium hydroxide and aqueous ammonia to form their neutral precipitates
 - $[\text{M}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) \rightarrow [\text{M}(\text{H}_2\text{O})_4(\text{OH})_2](\text{s}) + 2\text{H}_2\text{O}(\text{l})$
 - This type of reaction can be described as a precipitation reaction as a precipitate is formed

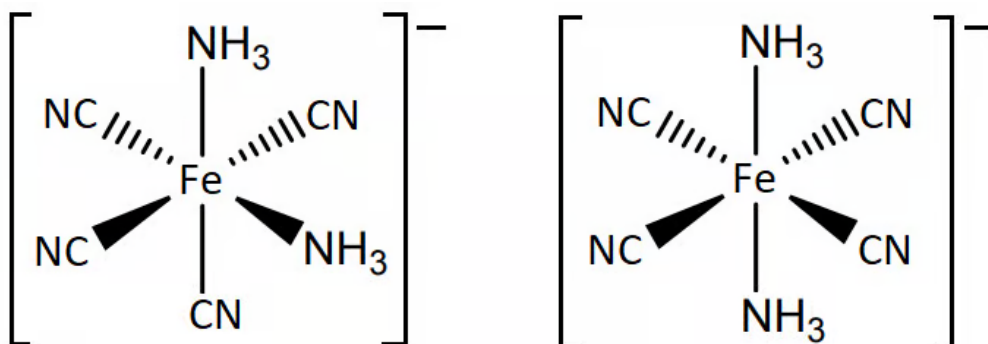
OR

It can be described as an acid-base reaction because the complex ion is behaving as an acid reacting with the sodium hydroxide base to form a neutral solution

- You need to know the colours associated with each reaction
 - The colour associated with iron(II) is green
- Only specific M^{2+} complex ions react with an excess of aqueous ammonia - Cu(II) and Co(II)
 - $[M(H_2O)_6]^{2+} + 6NH_3 \rightarrow [M(NH_3)_6]^{2+} + 6H_2O$
 - This type of reaction is described as ligand exchange because all of the ligands have been exchanged
 - In the specific case of Co(II), you may see some mark schemes describe the final appearance as a blue solution
 - This is because the $[Co(NH_3)_6]^{2+}$ easily oxidises in air to form $[Co(NH_3)_6]^{3+}$

2c

c) The three-dimensional diagrams showing the two isomers of $[Fe(NH_3)_2CN_3]^-$ are:



- Cis isomer showing 2 NH_3 groups at 90° to each other inside square brackets
AND
1– charge on the complex; [1 mark]
- Trans isomer showing 2 NH_3 groups at 180° to each other inside square brackets
AND
1– charge on the complex; [1 mark]

The type of stereoisomerism is:

- Geometric

OR

Cis / trans; [1 mark]

[Total: 3 marks]

- Do not be put off by the unfamiliar complex ion
- There are 6 ligands
 - Two are ammonia ligands
 - This means that they can either be next to each other forming a 90° angle OR they can be opposite each other forming a 180° angle
- This is geometric isomerism but the term cis / trans is accepted

2d

d)

i) One molecule of **A** can form three dative bonds because:

- Each nitrogen has a lone pair of electrons

OR

There are 3 lone pairs of electrons; [1 mark]

ii) The equation for the reaction of $\text{C}_4\text{H}_{13}\text{N}_3$ with aqueous chromium(III) ions, $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$, in a 2:1 ratio to form a new complex ion is:

- $[\text{Cr}(\text{H}_2\text{O})_6]^{3+} + 2\text{C}_4\text{H}_{13}\text{N}_3 \rightarrow [\text{Cr}(\text{C}_4\text{H}_{13}\text{N}_3)_2]^{3+} + 6\text{H}_2\text{O}$; [1 mark]

[Total: 2 marks]

- A ligand is a molecule / ion that has one or more lone pairs of electrons that can be donated to a central metal atom / ion forming dative covalent bonds in the process
- Since each compound **A** can form three dative covalent bonds, it will require two molecules of compound **A** to exchange with all six water ligands
 - $[\text{Cr}(\text{H}_2\text{O})_6]^{3+} + 2\text{C}_4\text{H}_{13}\text{N}_3 \rightarrow [\text{Cr}(\text{C}_4\text{H}_{13}\text{N}_3)_2]^{3+} + 6\text{H}_2\text{O}$
 - This agrees with the 2:1 ratio given in the question

3a

a) The electronic configurations of a Co atom and a Co^{2+} ion are:

- Co atom = $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$

OR

Co atom = $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$; [1 mark]

- Co^{2+} ion = $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$; [1 mark]

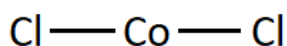
[Total: 2 marks]

- You should be aware that:
 - The Periodic Table makes it appear that electrons fill orbitals in the order 1s 2s 2p 3s 3p 4s 3d
 - However, electrons actually fill the orbitals in the order 1s 2s 2p 3s 3p 3d 4s
 - The 3d orbital comes before the 4s orbital because it is slightly lower in energy
 - s-orbitals can hold up to 2 electrons
 - p-orbitals can hold up to 6 electrons
 - d-orbitals can hold up to 10 electrons
- With 27 electrons, an atom cobalt will be $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$
 - You can check this by just focussing on the numbers: $2 + 2 + 6 + 2 + 6 + 7 + 2 = 27$
- A Co^{2+} ion has lost 2 electrons, which means that it has 25 electrons
 - **Remember:** 4s electrons are lost before 3d electrons
 - Since the 4s shell electrons are lost first, this will give $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$

- A Co^{2+} ion has lost 2 electrons, which means that it has 25 electrons
 - **Remember:** 4s electrons are lost before 3d electrons
 - Since the 4s shell electrons are lost first, this will give $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$

3b

b) The structure of the linear complex formed by Co^{2+} ions and Cl^- ions is:



- $\text{Cl} - \text{Co} - \text{Cl}$; [1 mark]

The overall charge of this complex is:

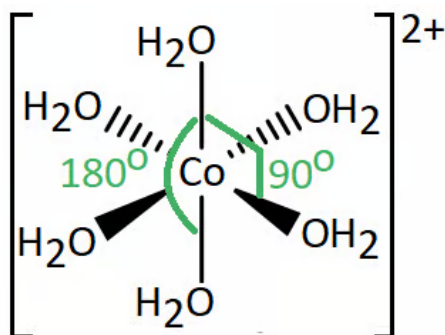
- Zero / 0 / no charge; [1 mark]

[Total: 2 marks]

- The question tells you that the complex is linear
- This means that the Co^{2+} ion will have two of the monodentate chloride ligands / ions arranged in a linear fashion, i.e. with a 180° bond angle
- One Co^{2+} ion with two Cl^- ions means that there will be no overall charge

3c

c) The three-dimensional diagram of a $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ complex ion including two bond angles is:



- Correct structure of the complex ion
AND
Correct 2+ charge; [1 mark]

The shape of the complex is:

- Octahedral; [1 mark]

The bond angles are:

- 90° correctly labelled on the diagram
AND
 180° correctly labelled on the diagram; [1 mark]

[Total: 3 marks]

- Complexes with 6 ligands form an octahedral shape
- The bond angles within an octahedral structure are 90° for adjacent ligands and 180° for opposite ligands
- Since water is a neutral ligand, the complex will have an overall charge that is equal to the charge of the metal ion, i.e. $2+$

3d

d) The observations and equations for reactions 1 and 2 are:

Reaction 1

- Observation = blue / blue-green precipitate; [1 mark]
- Equation = $[\text{Co}(\text{H}_2\text{O})_6]^{2+} + 2\text{OH}^- \rightleftharpoons \text{Co}(\text{H}_2\text{O})_4(\text{OH})_2 + 2\text{H}_2\text{O}$
OR
 $\text{Co}(\text{OH})_2 + 2\text{OH}^- \rightleftharpoons \text{Co}(\text{OH})_2 + 2\text{H}_2\text{O}$; [1 mark]

Reaction 2

- Observation = (straw) yellow / brown solution; [1 mark]
- Equation = $\text{Co}(\text{H}_2\text{O})_4(\text{OH})_2 + 6\text{NH}_3 \rightleftharpoons [\text{Co}(\text{NH}_3)_6]^{2+} + 2\text{H}_2\text{O} + 2\text{OH}^-$
OR
 $\text{Co}(\text{OH})_2 + 6\text{NH}_3 \rightleftharpoons [\text{Co}(\text{NH}_3)_6]^{2+} + 2\text{H}_2\text{O} + 2\text{OH}^-$; [1 mark]

[Total: 4 marks]

- Cobalt(II) and copper(II) are the two transition metals that undergo ligand substitution with aqueous ammonia

- You need to be aware that cobalt(II) undergoes full substitution with 6 ligands being replaced by ammonia
 - Copper(II) undergoes partial ligand substitution with only four ligands being replaced by ammonia
- You also need to be aware of the colours / states associated with these complexes and reactions
- You could also be asked a follow on question about the $[\text{Co}(\text{NH}_3)_6]^{2+}$ as it will readily oxidise from a straw yellow / brown solution to form a blue solution of $[\text{Co}(\text{NH}_3)_6]^{3+}$

3e

e)

i) A hexadentate ligand is:

- A ligand that can donate six lone pairs of electrons (to a central metal atom / ion); [1 mark]
- To form six dative (covalent) / coordinate bonds; [1 mark]

ii) The type of reaction is:

- Ligand exchange / displacement / replacement / substitution; [1 mark]

iii) The stability constant, K_{stab} , expression of $[\text{CoEDTA}]^{2-}$ in this reaction is:

$$K_{\text{stab}} = \frac{[\text{CoEDTA}]^{2-}}{[\text{Co}(\text{H}_2\text{O})_6]^{2+} [\text{EDTA}]^{4-}}; [1 \text{ mark}]$$

iv) The numerical value of the K_{stab} tells us that the $[\text{CoEDTA}]^{2-}$ complex ion is:

- (Very) stable

OR

More stable than $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$; [1 mark]

[Total: 5 marks]

- The term ligand refers to a chemical that is able to donate lone pairs of electrons to a central metal atom / ion
 - You will most commonly be asked about mono- and bidentate ligands
- Since all of the water ligands are replaced by one EDTA ligand, this is a ligand

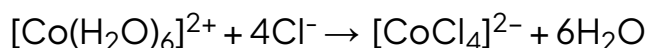
exchange reaction

- The equation for the reaction is:
 - $[\text{Co}(\text{H}_2\text{O})_6]^{2+} + \text{EDTA}^{4-} \rightleftharpoons [\text{CoEDTA}]^{2-} + 6\text{H}_2\text{O}$
- The reaction equation can be used (much like K_c) to deduce the K_{stab} expression
 - Remember:** Water does not feature in K_{stab} expressions
 - $K_{\text{stab}} = \frac{\text{products}}{\text{reactants}} = \frac{[[\text{CoEDTA}]^{2-}]}{[[\text{Co}(\text{H}_2\text{O})_6]^{2+}][[\text{EDTA}]^{4-}]}$
- Remember:** The greater the value of K_{stab} the more stable the complex is

4a

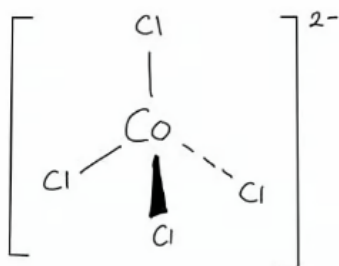
a)

i) The equation for the reaction when concentrated hydrochloric acid is added to $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$:



- $[\text{CoCl}_4]^{2-}$ complex including correct charge; [1 mark]
- Rest of the equation fully correct; [1 mark]

ii) The three-dimensional diagram to show the complex formed in the reaction between $[\text{Co}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$ and HCl is:



- Correct structure
AND
Correct charge; [1 mark]
- 3-D bonds used correctly
AND
Showing the tetrahedral shape of the complex; [1 mark]

[Total: 4 marks]

- In the ligand substitution reaction, the six water ligands are replaced by four chloride ions
- The shape of the complex formed is tetrahedral as it has four bonding pairs of electrons around the central metal ion
- As the chloride ion is larger than water a change in coordination number is observed

4b

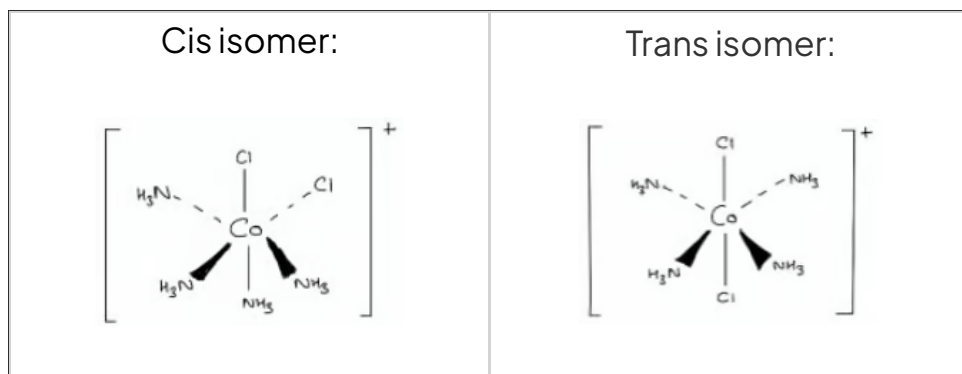
b)

i) The oxidation state of cobalt is:

- +3; [1 mark]
- The NH_3 ligands are neutral and the Cl^- ligands each contribute -1

ORThe NH_3 ligands are neutral and the Cl^- ligands overall contribute -2; [1 mark]

ii) The structures of each isomer are:



- Six correct ligands around Co

ANDBonds are shown from the N of NH_3 ; [1 mark]

- 3-D bonds used correctly for an octahedral structure; [1 mark]
- cis and trans isomers correctly identified; [1 mark]

[Total: 5 marks]

- In the cis isomer, the two Cl^- ligands are at 90° to each other / next to each other
- In the trans isomer, the two Cl^- ligands are at 180° to each other / opposite or across from each other

4c

c) Explanation of the origin of colour in a transition element complex such as $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$:

- Complexes have two sets of d orbital(s) of different energy

OR

d-orbitals split into two sets (of orbitals); [1 mark]

- Visible light absorbed (and complementary colour observed); [1 mark]
- Electron(s) are promoted / excited

OR

Electron(s) moves to higher (d–) orbital; [1 mark]

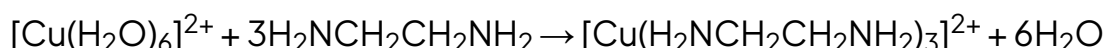
[Total: 3 marks]

- Questions about the origins of colour in transition metal complexes are typically for these same three marks
- So, it is worth taking the time to learn these points

4d

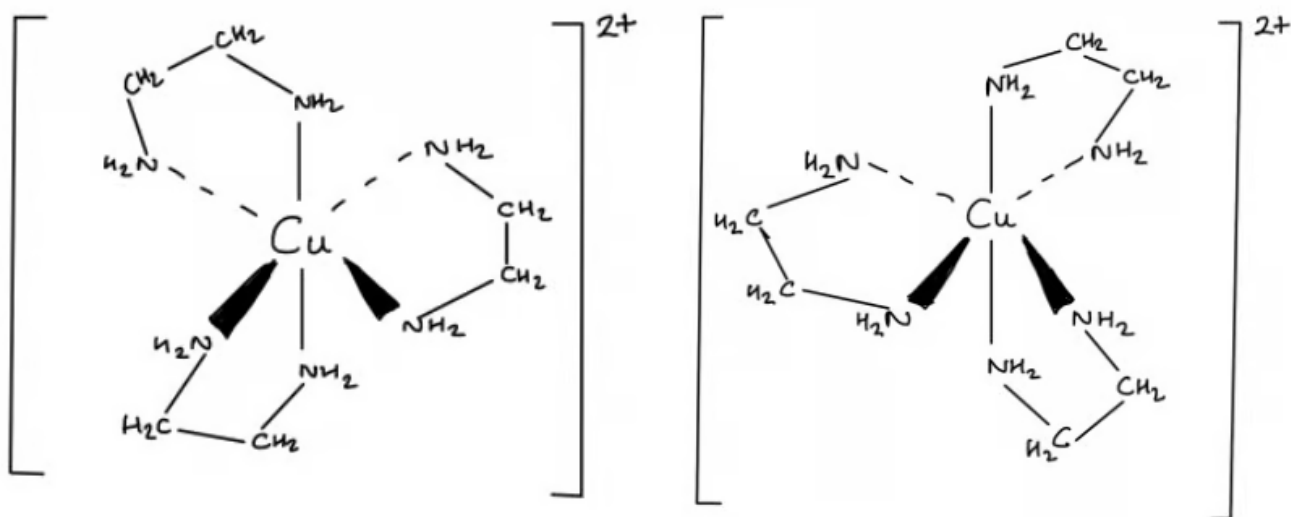
d)

i) The equation for the reaction taking place when $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ reacts with 1,2–diaminoethane is:



- $[\text{Cu}(\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2)_3]^{2+}$ complex correct; [1 mark]
- Rest of the equation fully correct; [1 mark]

ii) The displayed structures of the two isomers produced are:



- Three correct ligands around Cu

AND

Bonds are shown from the N of the 1,2-diaminoethane ligand; [1 mark]

- 3-D bonds used correctly to show an octahedral shape; [1 mark]
- Two mirror images to show optical isomerism; [1 mark]

[Total: 5 marks]

- Each of the 1,2-diaminoethane ligands can form 2 coordinate bonds to the central metal ion, as they are bidentate ligands
 - So only 3 molecules of 1,2-diaminoethane are required to replace all 6 of the water ligands